P&ID of Vinyl-Chloride Process

Mini Project 3

Che 396 - Senior Design I

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Contents

Project Charter

Title	Goal	In Scope	Out of Scope
P&ID of Vinyl Chloride Process	Design a control system for before and after heat integration by using appropriate materials of construction	● Design of the control system ● List of the materials of construction ● Handling alarms and conceptual system	● Performing heat integration ● Adding additional valves after heat integration

Deliverables	Team Members
Design Plan Material of Construction Control System P&ID Before and After Heat Integration	Yier Li Ariel Maret Tianna Mitchell Erick Yspango

Design Plan

Planing			October								November		
Execution													
Task		Assignee	24th	25th	26th	27th	28th	29th	30th	31st	1st	2nd	Days
Project Charter		All											2
Design Plan		All											2
Material of Contruction		All											5
Conceptual Control System	B: stream 1-7	Yier											6
	B: stream 8-16	Erik											
	A: valve 1-6	Tianna											
	A: valve 7-16	Ariel											
Project Flow Diagram		All											3
Layout / Formate		All											2
Section Unification		All											2
Proofread/Plagiarism Check		All											2

Objective

The objective of this project is to design a conceptual control system for before and after heat integration for a vinyl chloride process occurring via direct chlorination. Appropriate material of construction for the equipment and pipes will be provided for the system. For this vinyl chloride process, three control objectives are listed. There needs to be nearly 100% conversion achieved in the dichloroethane reactor, the conversion of the pyrolysis furnace cannot be altered and the control of the ethylene feed flow will be maintaining a production level. Control elements will be added and evaluated both before and after heat integration. It is observed that the process is continuous. The following topics will also be addressed: potential disturbances, constraints, safe limits for process variables, the decision of feedback or feedforward control, process sensor and final control element.

Materials of Construction

The following table was generated from a chemical compatibility chart provided with a temperature of 21°C. The options are listed in order of higher compatibility options with an * annotating the same compatibility grade of A [1].

Table 1. The general material Options for the following Components.

Component	Piping Material Options
Chlorine feed (vapor)	Fluorocarbon*, Hastelloy C*, PTFE*, 316 stainless steel
Ethylene feed (vapor)	Aluminum*, Cast/Ductile Iron*, 304 and 316 stainless steel*
1,2-dichloroethane (liquid)	Carbon steel*, Cast/Ductile Iron, 304 and 316 stainless steel
Hydrogen chloride (vapor)	304 and 316 stainless steel*, Cast/Ductile Iron
Hydrogen chloride (liquid)	Cast/Ductile Iron or 316 stainless steel
Vinyl Chloride	316 stainless steel*, 304 stainless steel, Carbon steel, Cast/Ductile Iron

Table 2. Material of Construction for the Units of the Process Before/After Integration [3, 6, 9, 10].

Unit (Before)	Unit (After)	Material
Direct Chlorination Reactor, R-100	Direct Chlorination Reactor, R-100	Stainless Steel
Reactor Condenser, E-100	Reactor Condenser, E-100	Stainless Steel
Reactor Pump, P-100	Reactor Pump, P-100	Cast Iron
Evaporator, E-101	HEN 1 (furnace and quench), E-101	Stainless Steel
Pyrolysis Furnace, F-100	Pyrolysis Furnace, F-100	Stainless Steel
Quench Tank, V-100	Quench Tank, V-100	Stainless Steel
Quench Pump, P-101	Quench Pump, P-101	Stainless Steel
Quench Cooler, E-102	Cooler, E-102	Stainless Steel
Recycle Cooler, E-108	Recycle Cooler, E-108	Stainless Steel
HCl Column, T-100	HCl Column, T-100	Stainless Steel
HCl Column Reboiler, E-105	HCl Column Reboiler, E-105	Stainless Steel
HCl Column Condenser, E-104	HCl Column Condenser, E-103	Stainless Steel
HCl Column Reflux Drum, V-101	HCl Column Reflux Drum, V-101	Carbon Steel
Condenser, E-103	HEN 2 (HCl column & Cooler), E-104	Stainless Steel
HCl Column Reflux Pump, P-102	HCl Column Reflux Pump, P-102	Cast Iron
VC Column, T-101	VC Column, T-101	Stainless Steel
VC Column Condenser, E-106	VC Column Condenser, E-106	Stainless Steel
VC Column Reboiler, E-107	VC Column Reboiler/HEN 3, E-107	Stainless Steel
VC Column Reflux Drum, V-102	VC Column Reflux Drum, V-102	Carbon Steel
VC Column Reflux Pump, P-103	VC Column Reflux Pump, P-103	Cast Iron
Recycle Pump, P-104	Recycle Pump, P-104	Cast Iron

Materials of Construction for the Piping of Streams 1-9

Streams 1-9 are the same for both before and after heat integration. The piping materials were chosen appropriately per stream depending on the parameters. The piping for the ethylene feed will consist of 316 stainless steel due to the highest compatibility as shown in **Table 1.** The chlorine feed pipe will consist of Hastelloy due to being the best for very reactive chemicals and having the highest compatibility [9]. The temperature of the reactor outlet raises to 90°C and consists of 1,2-dichloroethane. With the compatibility being the highest for carbon steel, which is also cheaper, carbon steel will be the piping material chosen for streams three through six. The furnace inlet stream of 1,2-dichloroethane raised to 242°C. The pipe for that stream will be carbon steel because it's cheap and it fits within the temperature range of carbon steel being -45°C to 480°C [9]. The outlet stream of that furnace consists of 1,2-dichloroethane, vinyl chloride and hydrogen chloride at 500°C. Since 316 stainless steel is compatible with all three components and ranges from -225°C to 1100°C, it will be chosen for stream 7, 8 and 9 also [9].

Table 3. Chosen Material of Construction for Streams 1-9 (Before and After Heat Integration)

Stream	**Material of Construction**
Ethylene Feed (1)	316 Stainless Steel
Chlorine Feed (2)	Hastelloy
Reactor Outlet (3)	Carbon Steel
Reactor Outlet + Recycle (4)	Carbon Steel
Evaporator/HEN 1 Inlet (5)	Carbon Steel
Furnace Inlet (6)	Carbon Steel
Furnace Outlet (7)	316 Stainless Steel
Quench Tank Outlet (8)	316 Stainless Steel
Condenser Outlet (9)	316 Stainless Steel

Before Heat Exchange Network

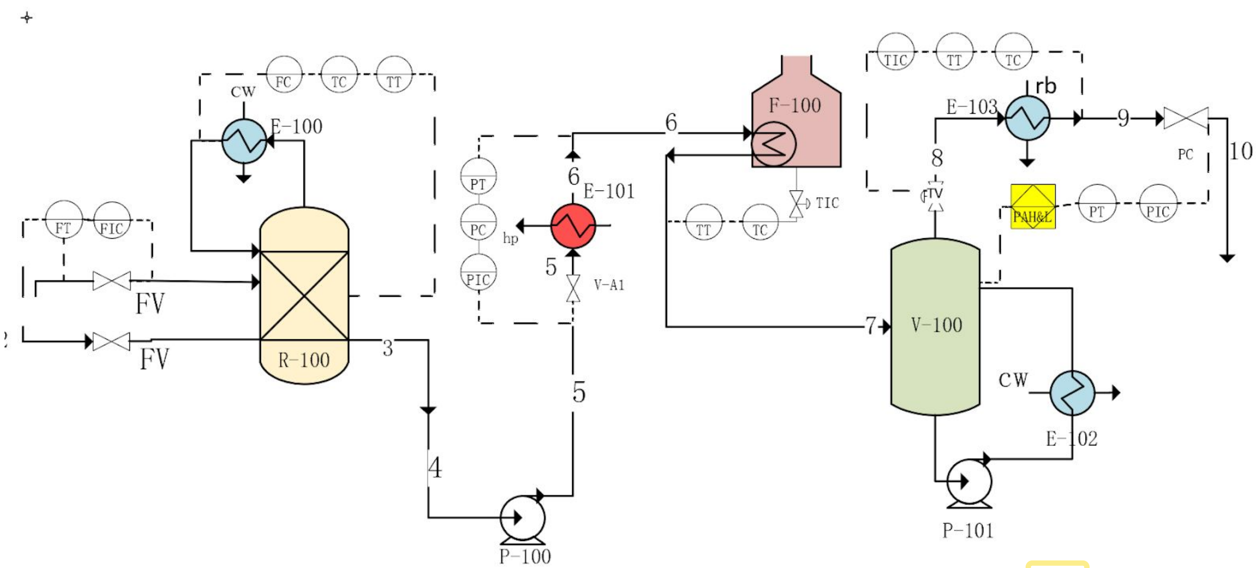

Figure 1. First Half of Before Heat Integration Process (stream 1 to stream 9)

Direct Chlorination Reactor (R100)

As listed in **Table 2** above, the best choice of materials for the direct chlorination reactor is stainless steel.

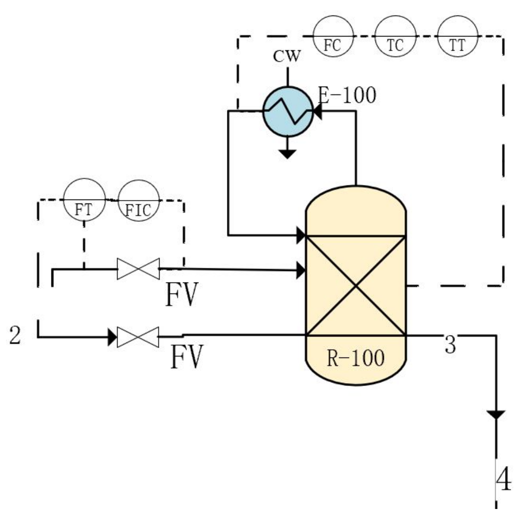

Figure 2. P&ID for Direct Chlorination Reactor

Table 4. List of control and manipulated variables and potential disturbances for reactor

Units	Controlled variables	Manipulated variable	Potential disturbance
Reactor (R 100)	Temperature	Flow rate	Temperature
	Pressure		pressure

Consider the reactor for the production of vinyl chloride shown in **Figure 2**. To control the objective is to keep the process safe and maintain the in and out production rate. In order to do this, there are several variables that need to be controlled. The controlled variables for the reactor unit are temperature, pressure, and level. The first part of the control is reactor feed, Ethylene, and chloride (stream 1 & 2) are fed in 44,900 and 113,400 lb/hr at 25 °C and 1.5 atm with a ratio between these two streams are determined. Two feed flows are controlled by a flow rate control system so the ratio of two-stream could always be constant. In order to regulate the temperature in the system, the direct chlorination reactor, is to control the flow rate of the cooler. So the temperature transmitter and flow rate control is added on the tube side of the heat exchanger and reactor. There will be no safety alarm set it up in the reactor unit.

Reactor Furnace (F-101)

As listed in **Table 2** above, the best choice of material for the pyrolysis furnace unit is stainless steel.

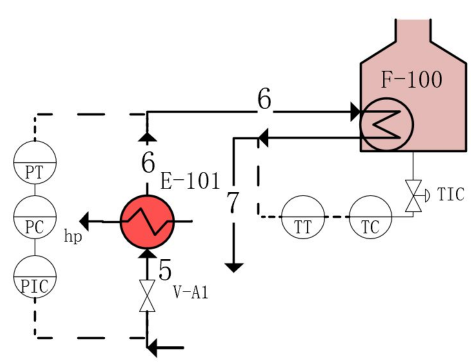

Figure 3. P&ID for Reactor Furnace

Table 5. the list of control and manipulated variables and potential disturbances for heater

Units	Controlled variables	Manipulated variable	Potential disturbance
Heater (F-101)	Temperature	pressure	Temperature
	Effluent flow rate		pressure

When the temperature difference between a process fluid and the furnace target temperature is more than 100-150 °C, then preheating could be necessary to bring the temperature closer to the furnace [7]. The flowrate into pyrolysis furnace gas, due to the heating equipment E-101, has a pressure control system that is used to control the heater. In this unit session, the level control is not needed based on the temperature at 91-500 °C, which is higher than vinyl chloride's boiling points (-134°C) [5]. The level control is not needed for this section. A main component of the pyrolysis furnace is the heating system. The primary goal for the pyrolysis furnace controlled variable is the coil outlet control. This is the same as the heater (E-101) which we need to control the temperature of the target flow to make sure enough heat released into the pyrolysis furnace chamber. In this case, only the simplest temperature control loop to adjust a control valve TIC on the fuel supply line to the heater. This furnace is not a batch process. Even though the temperature is heating up at a very high point, a temperature alarm is not needed in the furnace case.

Quench Tank (V-100)

For the quench tank, it is recommended to use stainless steel, since it is low in cost and with long corrosion time. The best choice of piping through each equipment is carbon steel.

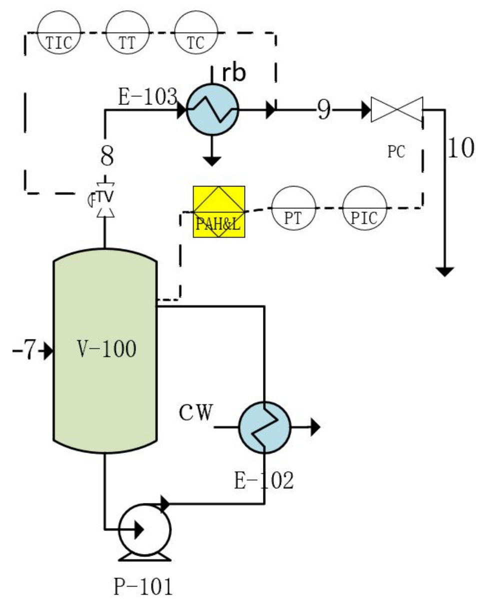

Figure 4. P&ID for Quench Tank

Table 6. List of control and manipulated variables and potential disturbances for heater

Units	Controlled variables	Manipulated variable	Potential disturbance
Heater (V-100)	Temperature	Pressure	Temperature
	Flow rate		Flow Rate

The Quench Tank is for process and storage purposes. The controlled variable for the Quench Tank is temperature and flow rate, since the level controlled is not needed for vapor-vapor phase. Between this equipment unit and the temperature cool down from cooler E-103, a pressure control system is required for the tank to help maintain a constant pressure at 26 atm. During this unit operation, it is known that the amount of mole flow rate and mass flow rate will stay the same on both inlet stream 7 and outlet stream 8. Since the pressure is controlled through the pressure control loop, the flow rate would not change in this condition. The potential disturbance is temperature and flow rate. This process is not a batch process. Therefore, the flow rate control loop would not be necessary in this loop. However, the safety alarm is needed for the Quench

Tank, which is at a constant 26 atm pressure. That pressure is high enough to cause an explosion. Pressure alarm high and low are shown in **Figure 4**.

Distillation Columns T-100 & T-101

In **Figure 5** below, before the heat integration process, is broken down into 2 distillation columns; (T-100 and T-101). As listed in **Table 2**, the material choice for the distillation columns is carbon steel. In **Figure 6,** Stream 10 enters the first distillation column (T-100) at a pressure of 12 atm. The pressure is maintained constant through streams 10-12. Since it is important to maintain constant pressure, a pressure valve was added in stream 10. For safety reasons, it is important to keep an eye on the levels of the columns. A level control indicator was added to the bottom distillate and an alarm will sound if levels are low. A temperature indicator and temperature control were added to E-104 (HCl Column Condenser) & E-105 (HCl column Rebolier). It is important to control the temperature going into V-101(HCl Column Reflux Drum) as well as the stream going back to the distillation column. Level controls were also added to V-101(HCl Column Reflux Drum). We do not want it to overflow or run dry.

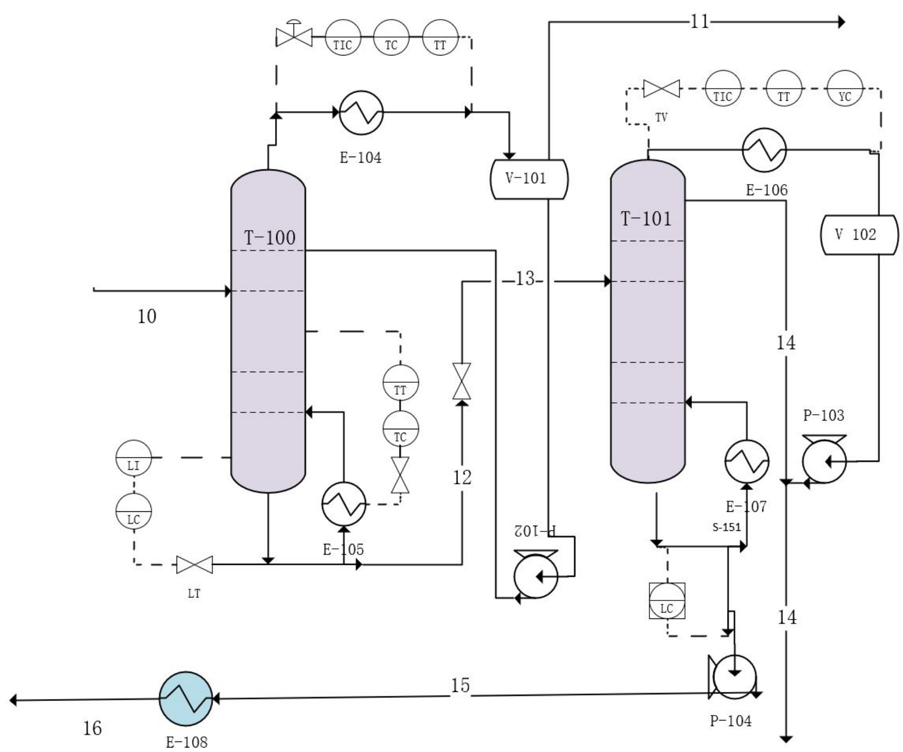

Figure 5. P&ID for Distillation Columns, T-100 and T-101

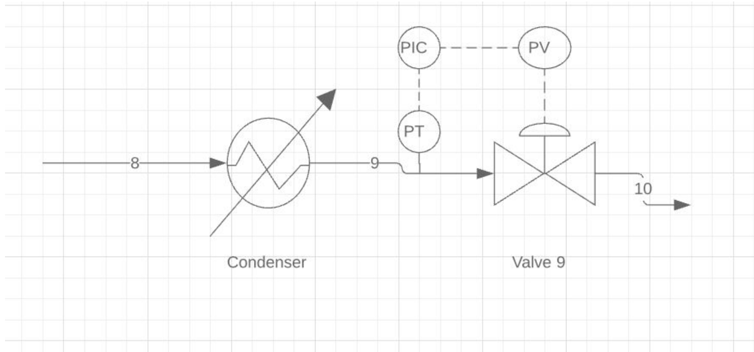

Figure 6 - P&ID for Valve 9

Table 7. Control System for Valve 9

V9	
Objective	Control the pressure of stream 10
Control Variable	Pressure stream 10
Manipulated Variable	Expansion Valve
Disturbance	None
Constraints	None
Feed Type	Feedback no immediate response required
Safety Limits	± 10% of the pressure. Range is (10.8 -13.2) atm.
Process Sensor	No Alarms
Final Control Element	Pressure Expansion Valve

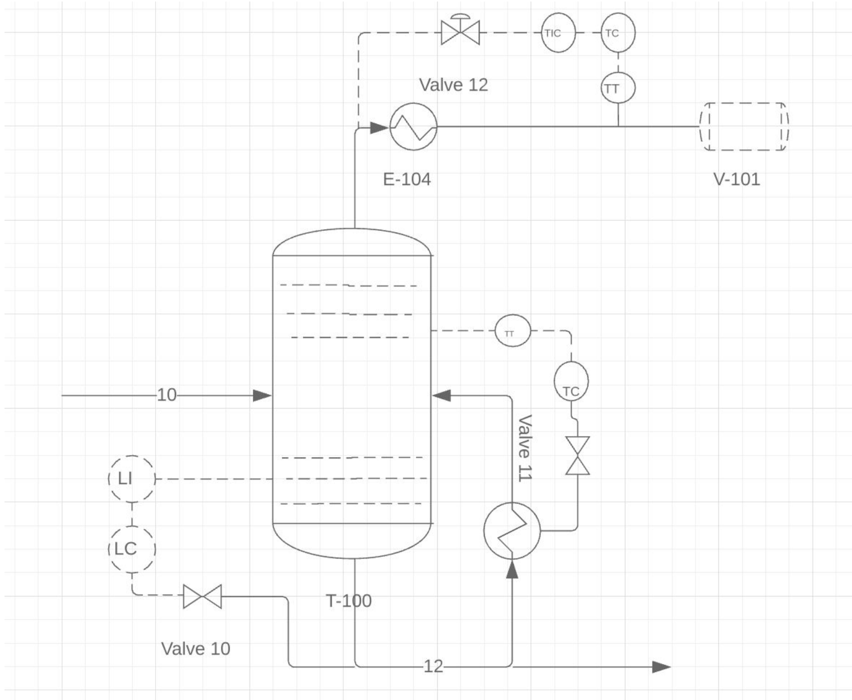

Figure 7 - P&ID for Valve 10-12

Table 8. Control System for Valve 10

V10	
Objective	Maintain proper levels in the bottoms
Control Variable	Liquid Stream
Manipulated Variable	Level Valve
Disturbance	None
Constraints	None
Feed Type	Feedback no immediate response required
Safety Limits	None
Process Sensor	Level Indicator
Final Control Element	Level Valve

Table 9. Control System for Valve 11

Objective	The objective of V11 is to control the reflux rate back into distillation column T-100.
Control Variable	The reflux flow rate back into the distillation column T-100.
Manipulated Variable	The liquid flow rate going back into the distillation column
Disturbance	A possible disturbance is if the stream is not all liquid. (Should be a liquid since a pump is next)
Constraints	The efficiency of the pump
Feed Type	Feed backward. No immediate response required.
Safety Limits	None
Process Sensor	No alarm needed
Final Control Element	Flow Control Valve

Table 10. Control System for Valve 12

Objective	Control the pressure of the stream going into the condenser E-103.
Control Variable	The pressure of the stream going into the condenser E-103.
Manipulated Variable	The flow of refrigerant into the partial condenser.
Disturbance	None
Constraints	None
Feed Type	Feed backward. No immediate response required.
Safety Limits	N/A
Process Sensor	No alarms needed.
Final Control Element	Pressure valve

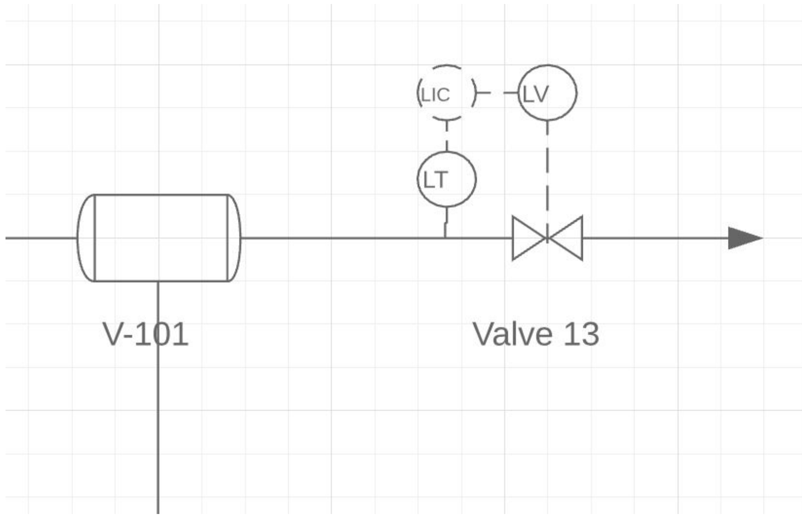

Figure 8 - P&ID for Valve 13

Table 11. Control System for Valve 13

V13	
Objective	Control the level of the reflux drum
Control Variable	The level of the reflux drum
Manipulated Variable	The vapor stream leaving the reflux drum (before the valve)
Disturbance	A possible disturbance is the pressure of the reflux drum
Feed Type	Feed backward. No immediate response required
Safety Limits	Height of the reflux drum, V-101
Process Sensor	None
Final Control Element	Level Control Valve

2nd Distillation Column- T101

Many of the steps described in T-100 are similar in T-101. **Figure 9** above shows the second distillation which is to begin with stream 13 entering the second distillation column (T-101) at a pressure of 4.8 atm. The pressure is maintained constant through streams 13-16. Since it is important to maintain a constant pressure, a pressure valve was added in stream 13. For safety reasons as mentioned earlier, it is important to keep an eye on the levels of the columns. A level control indicator was added to the bottom distillate and an alarm will sound if levels are low. A temperature indicator and temperature control was added to E-107 (VCl Column Reboiler) & E-106(VC Column Condenser).

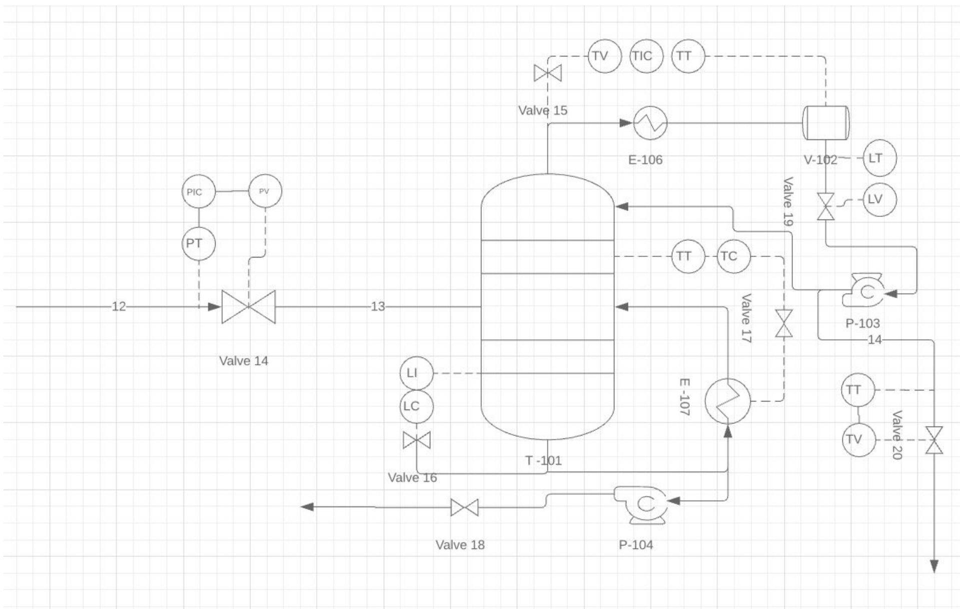

Figure 9- 2nd Distillation Column

Table 11. Control System for Valve 14

V14	
Objective	Control the pressure of stream 13
Control Variable	Pressure stream 13
Manipulated Variable	Expansion Valve
Disturbances	None
Constraints	None
Feed Type	Feedback no immediate response required
Safety Limits	± 10% of the pressure. Range is (10.8 -13.2) atm.
Process Sensor	No Alarms
Final Control Element	Pressure Expansion Valve

Table 10. Control System for Valve 15

Objective	Control the pressure of the stream going into the condenser E-106.
Control Variable	The pressure of the stream going into the condenser E-106.
Manipulated Variable	The flow of refrigerant into the partial condenser.
Disturbance	None
Constraints	None
Feed Type	Feed backward. No immediate response required.
Safety Limits	N/A
Process Sensor	No alarms needed.
Final Control Element	Pressure valve

Table 11. Control System for Valve 16

V16	
Objective	Maintain proper levels in the bottoms
Control Variable	Liquid Stream
Manipulated Variable	Level Valve
Disturbances	None
Constraints	None
Feed Type	Feedback no immediate response required
Safety Limits	None
Process Sensor	Level Indicator
Final Control Element	Level Valve

Table 12. Control System for Valve 17

Objective	The objective of V17 is to control the reflux rate back into distillation column T-101.
Control Variable	The reflux flow rate back into the distillation column T-101.
Manipulated Variable	The liquid flow rate going back into the distillation column
Disturbance	A possible disturbance is if the stream is not all liquid. (Should be a liquid since a pump is next)
Constraints	The efficiency of the pump
Feed Type	Feed backward. No immediate response required.
Safety Limits	None
Process Sensor	No alarm needed
Final Control Element	Flow control valve

Table 13. Control System for Valve 18

Objective	The objective of V18 is to control the flow rate of the stream going into condenser E108
Control Variable	The flow rate of the stream going into condenser E108
Manipulated Variable	The flow rate leaving pump P104 before valve 18
Disturbance	Boil-up ratio
Constraints	The efficiency of the pump
Feed Type	Feed backward. No immediate response required.
Safety Limits	None
Process Sensor	None
Final Control Element	Flow Valve

Table 14. Control System for Valve 19

Objective	Control the level of the reflux drum
Control Variable	The level of the reflux drum
Manipulated Variable	The vapor stream leaving the reflux drum (before the valve)
Disturbance	A possible disturbance is the pressure of the reflux drum
Constraints	Need a minimum amount of vapor in the reflux drum.
Feed Type	Feed backward. No immediate response required.
Safety Limits	Height of the reflux drum V102
Process Sensor	None
Final Control Element	Level Control Valve

`**Table 15. Control System for Valve 20**

Objective	The objective of V20 is to control the temperature of the top of distillation column T-101
Control Variable	The temperature of the top of distillation column T-101
Manipulated Variable	The flow rate of stream 14
Disturbance	The ambient temperature
Constraints	Must reach the bubble point temperature for vinyl chloride for it to become a vapor
Feed Type	Feed backward. No immediate response required.
Safety Limits	None
Process Sensor	Temperature Alarm High and Alarm Low
Final Control Element	Temperature Valve

After Heat Integration

Valve 1 Through Valve 7

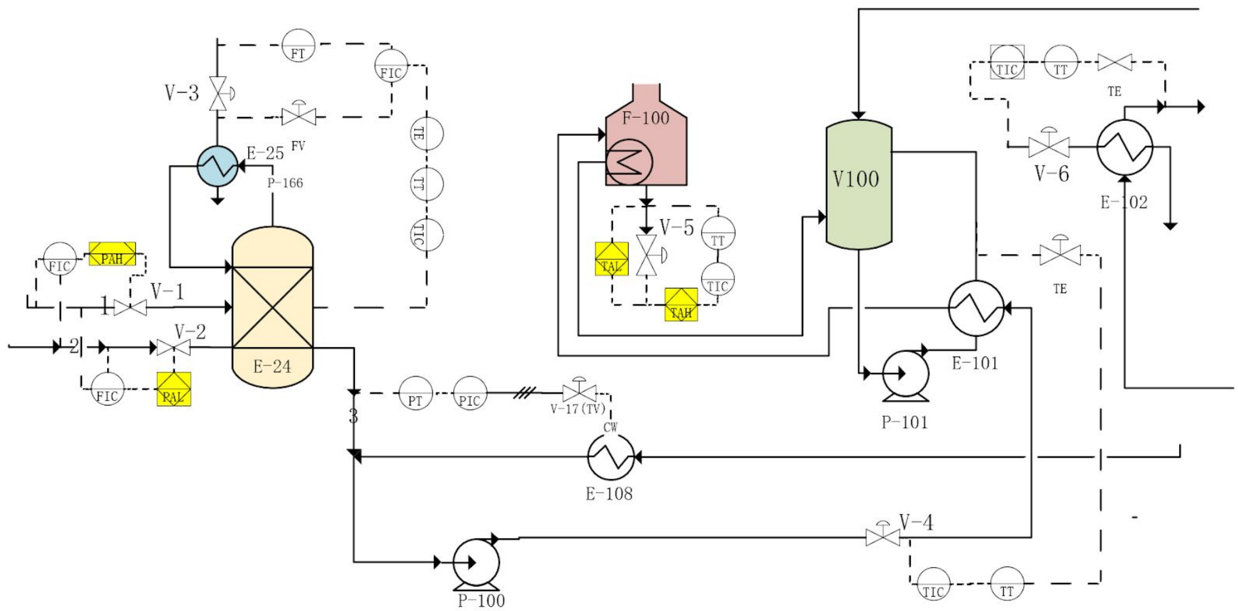

Figure 10. P&ID for First Half of Process (Valve 1 through 7)

Valve 1 corresponds to the feed of Ethylene into the direct chlorination reactor, R-100. The production level of the reactor is maintained by flow control of ethylene. Level control is important because it sets the inventory in process equipment and prevents accumulation from occurring in liquid phase reactors [8]. Low levels can cause damage to the equipment while high levels can cause an overflow. Since the reactor is not a liquid phase, a level control valve is not needed [3].

Table 16. Control System for Valve 1

V1	
Objective	The objective of the valve is to control the feed flow rate of ethylene
Control Variable	The control variable is the flow of ethylene.
Manipulated Variable	The manipulated variable is the valve for the feed flow.
Disturbances	Too high or too low of feed can result in undesired product

Constraints	N/A
Feed Type	Feedforward so the output is not affected
Safety Limits	±10% stability/repeatability
Process Sensor	High and low alarms
Final Control Element	Flow Control Valve

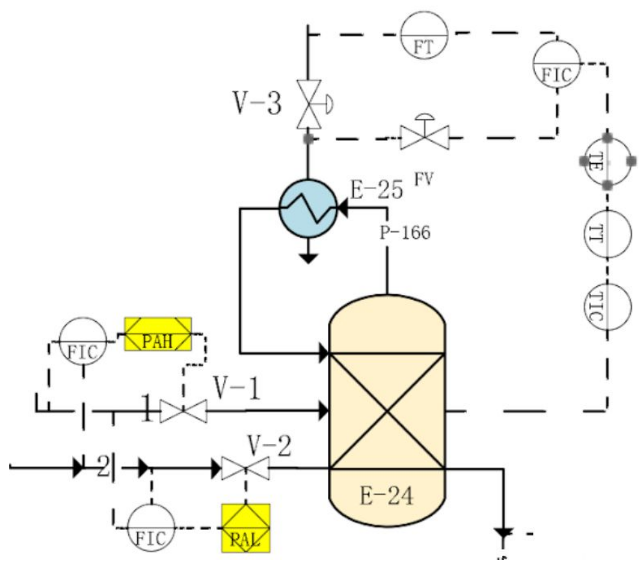

Figure 11. P&ID Around the Reactor (Valve 1-3)

Valve 2 corresponds to the feed of Chlorine into the direct chlorination reactor, R-100. The final control element on Valve 2 is a Flow Ratio Control Valve. The valve will be controlling the feed flow rate of chlorine in ratio to ethylene to maintain the proper stoichiometry for the reactor. The feed type chosen is a feedforward due to the measurement of the ethylene feed flow rate.

Table 17. Control System for Valve 2

V2	
Objective	The objective of the valve is to control the feed flow rate of chlorine in ratio to ethylene
Control Variable	The control variable is feed flow rate of ethylene

Manipulated Variable	The manipulated variable is the feed flow rate of chlorine
Disturbance	Feed flow of ethylene
Constraints	N/A
Feed Type	Feedforward because flow of ethylene needs to be measured
Safety Limits	N/A
Process Sensor	No alarms
Final Control Element	Flow Ratio Control Valve

Valve 3 is placed on a cold utility stream for the reactor condenser, E-100. By controlling the feed flow of the cold water utility, it will in return, control the temperature of the reactor. The feed flow controller would be the primary controller which would adjust the secondary controller, which is the temperature controller.

Table 18. Control System for Valve 3

V3	
Objective	The objective of the valve is to control the temperature of the reactor
Control Variable	The control variable is the temperature of the reactor
Manipulated Variable	The manipulated variable is the flow of the cold utility
Disturbances	Possible disturbance is inlet feed temperature
Constraints	N/A
Feed Type	Feedback because no immediate response is required
Safety Limits	Allows max capacity of 95% pressure control
Process Sensor	No alarms
Final Control Element	Flow Control Valve

Valve 4 is placed after the reactor pump, P-100, and before the quench cooler, E-101. The final control element will be a temperature control valve to control the temperature of the quench cooler. The quench cooler is being controlled because 1,2-dichloroethane, Vinyl Chloride and Hydrogen Chloride enter the quench tank, V-100, at 500°C and must be cooled to 170°C before leaving the tank. The solution needs to continuously go through cooling until the desired exit temperature is reached.

Table 19. Control System for Valve 4

V4	
Objective	The objective of the valve is to control the temperature of the quench cooler
Control Variable	The control variable is the temperature of the cooler
Manipulated Variable	The manipulated variable is the valve
Disturbance	If the feed has vapor in it
Constraints	Efficiency of pump
Feed Type	Feedback because no immediate response is required
Safety Limits	N/A
Process Sensor	No alarms
Final Control Element	Temperature Control Valve

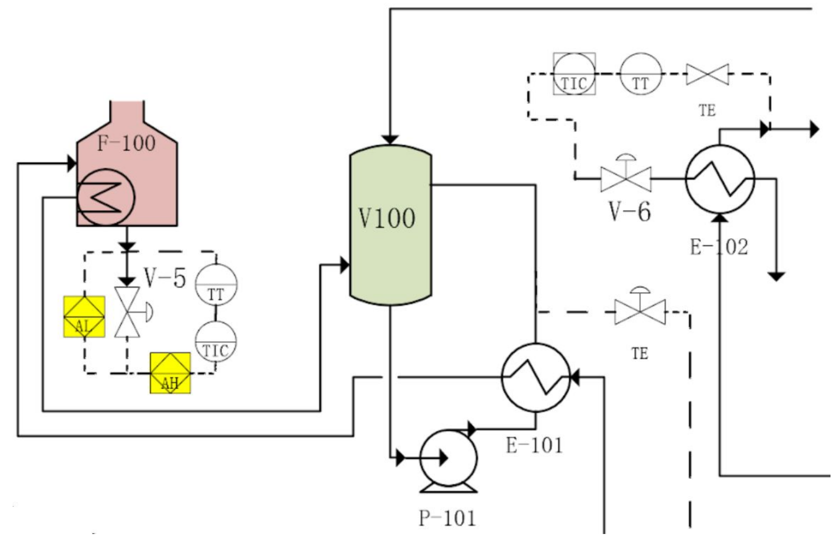

Figure 12. P&ID around Furnace and Quench Tank (Valve 5 and 6)

Valve 5 corresponds to the hot utility of fuel gas for the pyrolysis furnace, F-100. Due to the solution of 1,2-dichloroethane, Vinyl Chloride and Hydrogen chloride entering the pyrolysis furnace at 242°C and exiting at 500°C, the temperature needs to be controlled. If the temperature reaches too high or too low, an alarm needs to be set because it can alter the results.

Table 20. Control System for Valve 5

V5	
Objective	The objective of the valve is to control the temperature of the furnace
Control Variable	The control variable is the furnace temperature
Manipulated Variable	The manipulated variable is the valve of fuel gas
Disturbance	Temperature or flow of fuel gas
Constraint	Total heat released
Feed Type	Feedforward to ensure correct temperature range is implemented
Safety Limits	Fuel flow rate should be monitored
Process Sensor	Alarm high and low
Final Control Element	Temperature Control Valve

Valve 6 corresponds to the cold utility of refrigerated brine for a condenser, E-102, prior to the HCl column, T-100. The exit stream from the quench tank, V-100, goes through this condenser for further cooling of the stream before entering the HCl column.

Table 21. Control System for Valve 6

V6	
Objective	The objective of the valve is to control the temperature of cooler prior to the HCl column
Control Variable	The control variable is the temperature of the cooler
Manipulated Variable	The manipulated variable is the valve of refrigerated brine
Disturbances	N/A
Constraints	N/A
Feed Type	Feedback. No immediate response required.
Safety Limits	N/A
Process Sensor	No alarms
Final Control Element	Temperature Control Valve

The valve on the cold water utility on the recycle cooler, E-108, going back into the reactor will be annotated as V108. This valve will be controlled by a pressure controller because a temperature controller is less likely to be effective [3].

Table 22. Control System for V108

V108	
Objective	The objective of the valve is to control the temperature with a pressure controller
Control Variable	The control variable is the temperature of the cooler
Manipulated Variable	The manipulated variable is the pressure valve.

Disturbances	N/A
Constraints	N/A
Feed Type	Feedback. No immediate response required
Safety Limits	±10% of the pressure
Process Sensor	No alarms
Final Control Element	Pressure Control Valve

Materials of Construction for Streams 10-16

The material of construction depends on the material in the stream or vessel. There are three components, which are 1,2 dichloroethane, vinyl chloride, and HCl. See the table below for the possible materials for each component.

Table 23: Materials of Construction: After Heat Integration

Valve 7-Valve 16 [1]	
Component	**Possible Materials**
1,2 - dichloroethane	SS 302, SS 304, SS 316, Hastelloy, Inconel, Elgiloy
Vinyl Chloride	SS 302, SS 304, SS 316, Hastelloy, Inconel, Elgiloy
HCl	SS 316, Elgiloy

From **Table 23,** there are two choices for materials, which are SS 316 and Elgiloy. To determine which material should be used, the pros and cons for each material are evaluated.

For SS 316, the maximum operating temperature for the material is 287.77 °C. For the pumps, heat exchangers, and distillation columns the range of temperatures is -26.4°C to 170°C. This means that SS 316 will work for the processes. [2]

For Elgiloy, the maximum operating temperature for the material is 426.67°C which is also within the ranges of the process. This means that based on temperature the Elgiloy works for the process.

Since both materials will operate within the temperature range of the process, the next determining factor is the cost of each material. Unfortunately, data was not found for Elgiloy. However, since Eligiloy has a higher operating temperature, it is assumed to be more expensive than SS 316.

Valve 7 through Valve 16

Controllers were placed on the valves only, not on the heat exchangers or pumps. The reason is that the valves are either before or after these types of equipment. By controlling the valves, heat exchangers and pumps are indirectly controlled.

The goal of valve 7 is to control the pressure of stream 10. Stream 10 is the feed stream for distillation column T-100. In order to control the pressure of components coming into the distillation column, an expansion valve is needed.

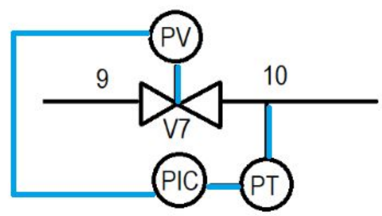

Figure 13: P&ID for Valve 7

Table 24. Control Systems for Valve 7

V7	
Objective	The objective of the expansion valve is to control the pressure of stream 10.
Control Variable	The control variable is the pressure of stream 10.
Manipulated Variable	The manipulated variable is the expansion valve
Disturbance	N/A
Feed Type	Feedback because no immediate response is required.
Safety Limits	± 10% of the pressure. Range is (10.8 -13.2) atm.
Process Sensor	No alarms
Final Control Element	Pressure Expansion Valve

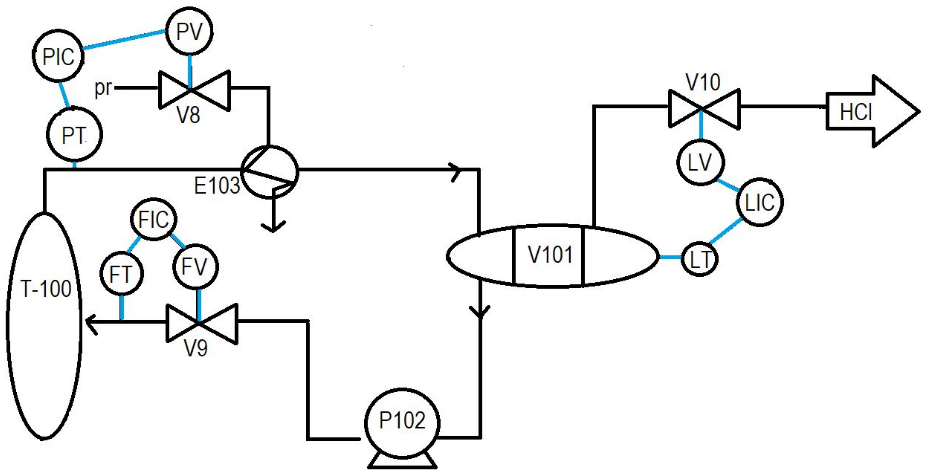

Figure 14: After Heat IntegrationValves 8, 9, and 10

For the first distillation column, T-100, the temperature of the bottom of the distillation column must be controlled. The pressure of the top distillate must be controlled as it goes into the condenser and the reflux rate going back into the distillation column must be controlled. Valve 8 is used to control the pressure of the stream going into the condenser E103. This occurs because the pressure in that stream is the same as the pressure in the column. Valve 9 is used to control the reflux rate of liquid back into the column. This allows the operator to control the purity. Valve 10 is used to control the level inside the reflux drum by changing the vapor stream leaving the reflux drum. This ensures that the drum does not overfill.

Table 25. Control System for Valve 8 and 9

	V8	V9
Objective	The objective of V8 is to control the pressure of the stream going into the condenser E-103.	The objective of V9 is to control the reflux rate back into distillation column T-100.
Control Variable	The pressure of the stream going into the condenser E-103.	The reflux flow rate back into the distillation column T-100.
Manipulated Variable	The flow of refrigerant into the partial condenser.	The liquid flow rate going back into the distillation column
Disturbance	None	A possible disturbance is if the stream is not all liquid. (Should be a liquid since a pump is next)
Constraints	None	The efficiency of the pump

Feed Type	Feed backward. No immediate response required.	Feed backward. No immediate response required.
Safety Limits	N/A	None
Process Sensor	No alarms needed.	No alarm needed
Final Control Element	Pressure valve	Flow control valve

Table 26. Control System for Valve 10

V10	
Objective	The objective of V10 is to control the level of the reflux drum
Control Variable	The level of the reflux drum
Manipulated Variable	The vapor stream leaving the reflux drum (before the valve)
Disturbance	A possible disturbance is the pressure of the reflux drum
Constraints	Need a minimum amount of vapor in the reflux drum.
Feed Type	Feed backward. No immediate response required.
Safety Limits	Height of the reflux drum V101
Process Sensor	None
Final Control Element	Level Control Valve

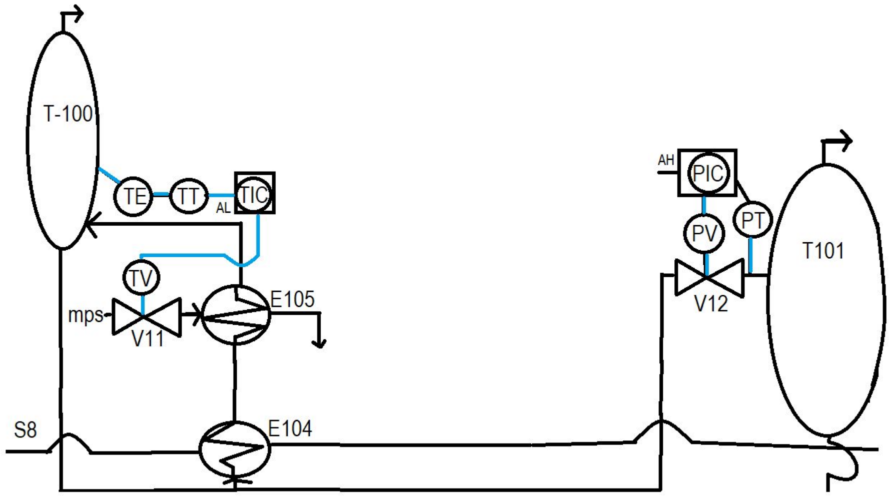

Figure 15: After Heat Integration Valves 11 and 12

The goal of valve 11 is to control the temperature of the bottom of the distillation column. The temperature control is important to have at the bottom of the column because the product vinyl chloride leaves the bottom of the column. The temperature determines how the different components will separate, so having control is very important. For this reason, a low alarm was chosen because if the temperature of the column is too low, HCl and vinyl chloride will not separate.

Table 27. Control System for Valve 11 and 12

V11		V12
Objective	The objective of V11 is to control the temperature of the bottom of the distillation column T-101	The objective of V12 is to control the pressure of the stream going back into distillation column T-101
Control Variable	Temperature the bottom of the distillation column	The pressure of the stream going into distillation column T-101
Manipulated Variable	The flow rate of the medium pressure steam	The expansion valve
Disturbance	A potential disturbance is a change in ambient temperature	A potential disturbance is the boil-up ratio
Constraint	Need a minimum flow rate of steam to ensure the temperature reached the Bubble Point	The bubble point pressure must be reached.

Feed Type	Feed backward. No immediate response required.	Feed backward. No immediate response required.
Safety Limits	N/A	± 10% of the pressure of stream 14. (4.32atm - 5.28atm)
Process Sensor	Temperature sensor - Low alarm needed	Pressure sensor Alarm High
Final Control Element	Temperature Valve	Pressure Expansion Valve

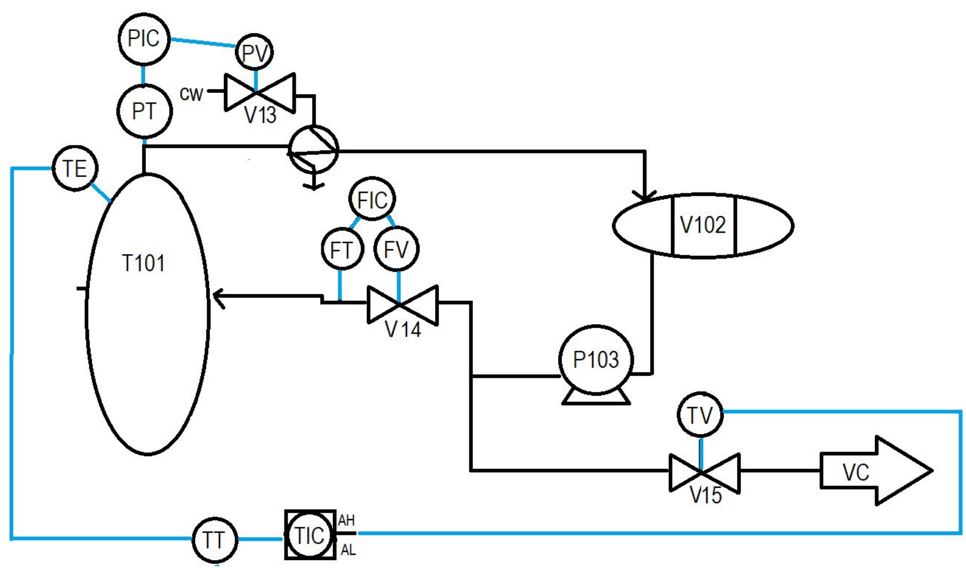

Figure 16: After Heat Integration Valves 13, 14, 15

The goal of valve 13 is to control the pressure on the top of distillation column T-101. The pressure of the stream leaving the top of the distillation column is the same as the pressure at the top of the distillation column. Valve 14 controls the reflux flow rate back into the distillation column. The reflux rate is important because it controls the purity of the product. Valve 15 is a temperature valve. Since our product from this column, vinyl chloride, leaves the top of the distillation column and the temperature sensor is placed at the top of the column. Since the temperature is critical for separation, an alarm high and alarm low is added. Both alarms are added because the temperature cannot be too low or too high in order for the separation to occur.

Table 28. Control System for Valves 13 and 14

	V13	V14
Objective	The objective of V13 is to control the pressure of the stream going into condenser E106.	The objective of V14 is to control the reflux rate from the drum into distillation column T-101
Control Variable	The pressure of the stream going into condenser E106.	Reflux rate from the drum into distillation column T-101
Manipulated Variable	The flow rate of the cooling water	The flow rate from the pump, P103, to valve 14
Disturbance	None	A potential disturbance is if there is not enough liquid.
Constraints	None	The efficiency of the pump
Feed Type	Feed backward. No immediate response required.	Feed backward. No immediate response required.
Safety Limits	N/A	None
Process Sensor	None	None
Final Control Element	Pressure Compression Valve	Flow Valve

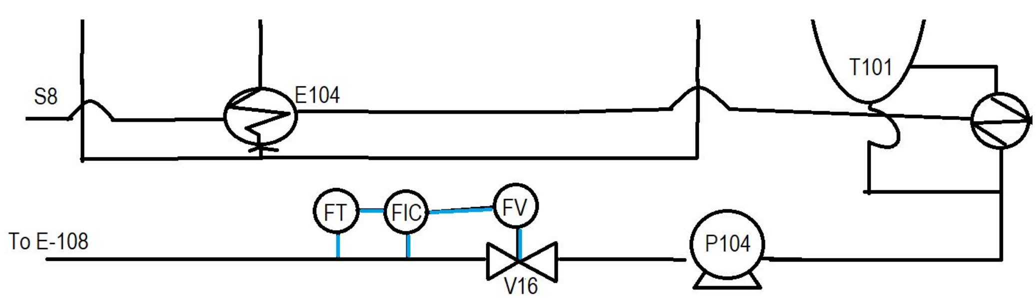

Figure 17: After Heat Integration: Valve 16 which controls the flow rate of the stream going into the condenser, E-108

Table 29. Control System for Valves 15 and 16

	V15	V16
Objective	The objective of V15 is to control the temperature of the top of distillation column T-101	The objective of V16 is to control the flow rate of the stream going into condenser E108
Control Variable	The temperature of the top of distillation column T-101	The flow rate of the stream going into condenser E108
Manipulated Variable	The flow rate of stream 14	The flow rate leaving pump P104 before valve 16
Disturbance	The ambient temperature	Boil-up ratio
Constraint	Must reach the bubble point temperature for vinyl chloride for it to become a vapor	The efficiency of the pump
Feed Type	Feed backward. No immediate response required.	Feed backward. No immediate response required.
Safety Limits	None	None
Process Sensor	Temperature Alarm High and Alarm Low	None
Final Control Element	Temperature Valve	Flow Valve

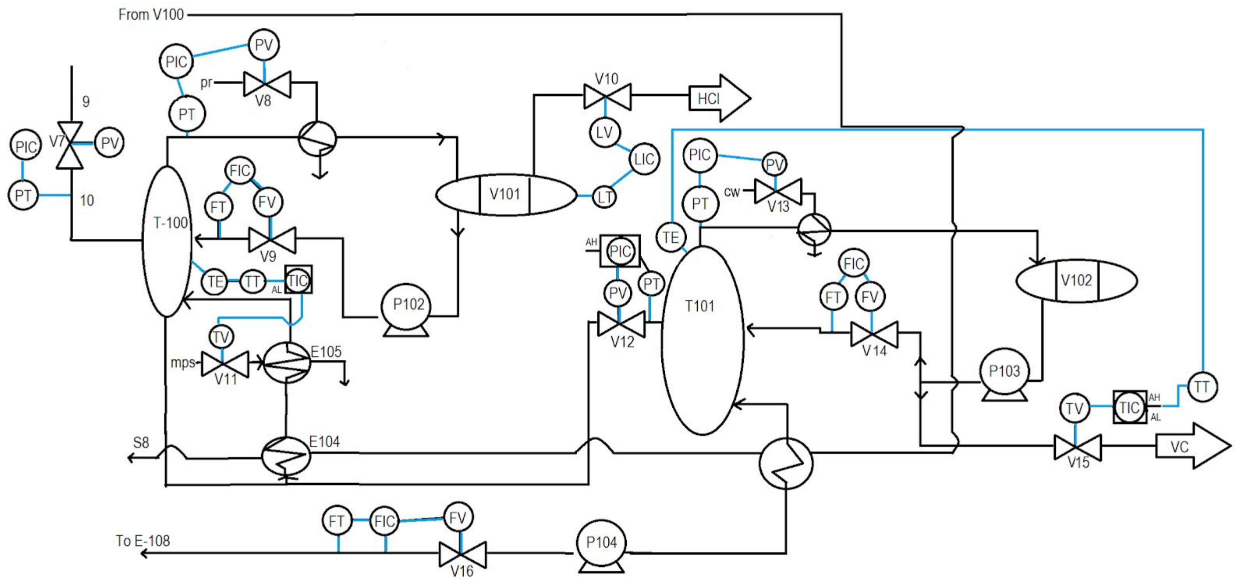

Figure 18: After Heat Integration Valves 7-16

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